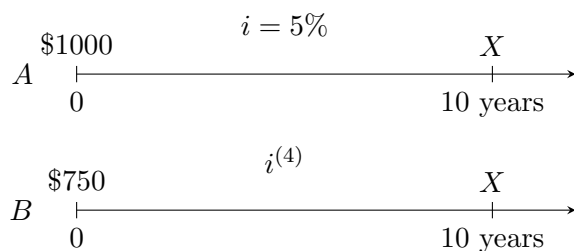


Problem 9.8

You deposit \$1,000 into Account A and \$750 into Account B. Account A earns an effective annual interest rate of 5%. Account B earns interest at $i^{(4)}$. Ten years later, the two accounts have the same accumulated value. Find $i^{(4)}$. Express as a percentage, to two decimal places.

Solution

We represent the two accounts on timelines:



For Account A, the accumulated value after 10 years is

$$1000(1.05)^{10}.$$

For Account B, since the nominal rate is convertible quarterly, there are $4(10) = 40$ conversion periods in 10 years. Its accumulated value is

$$750 \left(1 + \frac{i^{(4)}}{4} \right)^{40}.$$

Since the two accumulated values are equal,

$$1000(1.05)^{10} = 750 \left(1 + \frac{i^{(4)}}{4} \right)^{40}.$$

Solving for $i^{(4)}$,

$$\begin{aligned} \left(1 + \frac{i^{(4)}}{4} \right)^{40} &= \frac{1000(1.05)^{10}}{750}, \\ 1 + \frac{i^{(4)}}{4} &= \left(\frac{1000(1.05)^{10}}{750} \right)^{1/40}, \\ i^{(4)} &= 4 \left[\left(\frac{1000(1.05)^{10}}{750} \right)^{1/40} - 1 \right], \\ i^{(4)} &\approx 0.078315. \end{aligned}$$

Therefore,

$$i^{(4)} \approx 7.83\%.$$